

Bridge to IGCSE — Science & Physics Skills

WORKSHEET · TEST YOURSELF

For Grade 8 / Year 8 scientists making the jump from Checkpoint to IGCSE.

Answer on paper or below. **Show your working** and **give units**. Solutions are on the last page — try every question before you check.

A · Maths fluency

1. Rearrange $v = s/t$ to make s the subject, then to make t the subject.
2. Rearrange $p = m/V$ to make m the subject, then to make V the subject.
3. Convert: (a) 3.5 km to m (b) 250 g to kg (c) 4 min to s (d) 2 m³ to cm³ (e) 5000 cm³ to m³.
4. Write in standard form: (a) 6 200 000 (b) 0.0036.
5. Write as ordinary numbers: (a) 2.5×10^4 (b) 7×10^{-3} .
6. Round 0.04763 to 2 significant figures.
7. A spring extends 4 cm under a 2 N force (extension $\propto$ force). What is the extension under 5 N?
8. For a fixed mass of gas, pressure is inversely proportional to volume. If the volume halves, what happens to the pressure?
9. A student scores 36 out of 45. What percentage is this?
10. A car's speed rises from 20 m/s to 25 m/s. Calculate the percentage increase.

B · Command words

11. What does **explain** require that **describe** does not?
12. A question says "Calculate the speed." Name two things your answer must include besides the number.
13. "Compare metal X and metal Y." What must your answer contain to gain full marks?

C · Graphs & data

14. You change the temperature and measure a gas's volume. Which variable goes on the x-axis?
15. State the two things every graph axis must show.
16. A distance–time graph is a straight line from (0 s, 0 m) to (4 s, 20 m). Calculate the gradient and state what it represents.
17. What does the area under a speed–time graph represent?

D · Measuring & practical

18. What error are you avoiding by putting your eye level with a scale?
19. When reading a measuring cylinder, exactly where do you take the reading?
20. Three readings of a time are 4.1 s, 4.3 s and 4.2 s. Calculate the mean.
21. Give a one-sentence definition of **accuracy** and of **precision**.
22. A balance reads 0.2 g with nothing on it. Is this a random or a systematic error?

E · Planning an investigation

23. A student investigates how the **length of a wire** affects its **resistance**. Identify the independent variable, the dependent variable and one control variable.
24. What is the difference between a **result** and a **conclusion**?

F · Physics foundations

25. Why is "the mass is 5" not a complete answer in physics?
26. Sort these into scalars and vectors: speed, force, distance, velocity, mass.
27. A block has mass 240 g and volume 30 cm³. Calculate its density and give the unit.
28. A runner travels 100 m in 12.5 s. Calculate the speed to 2 s.f., with a unit.
29. Name two energy stores, and state the principle of conservation of energy in one sentence.

Worksheet — Solutions

FOR THE TEACHER / SELF-MARKING

For Grade 8 / Year 8 scientists making the jump from Checkpoint to IGCSE.

A · Maths fluency

1. $s = vt$; $t = s/v$.
2. $m = \rho V$; $V = m/\rho$.
3. (a) 3500 m (b) 0.25 kg (c) 240 s (d) 2 000 000 cm³ (e) 0.005 m³.
4. (a) 6.2×10^6 (b) 3.6×10^{-3} .
5. (a) 25 000 (b) 0.007.
6. 0.048.
7. 2 N → 4 cm means 2 cm per N, so 5 N → **10 cm**.
8. The pressure **doubles**.
9. $36 \div 45 \times 100 = \mathbf{80\%}$.
10. Change = 5 m/s; $5 \div 20 \times 100 = \mathbf{25\%}$.

B · Command words

11. **Explain** needs a reason — *why* it happens; **describe** only states what happens.
12. The **unit**, and the **working** (and a sensible number of significant figures).
13. Both **similarities AND differences**, linked between the two metals.

C · Graphs & data

14. Temperature — the independent variable goes on the **x-axis**.
15. The **quantity** measured and its **unit**.
16. Gradient = $20 \div 4 = \mathbf{5}$; it represents the **speed = 5 m/s**.
17. The **distance travelled**.

D · Measuring & practical

18. **Parallax** error.
19. At the **bottom of the meniscus**, with your eye level with it.
20. $(4.1 + 4.3 + 4.2) \div 3 = 12.6 \div 3 = \mathbf{4.2\ s}$.
21. **Accuracy**: how close a reading is to the true value. **Precision**: how close repeated readings are to each other.
22. A **systematic** error (a zero error).

E · Planning an investigation

23. Independent = **length of wire**; dependent = **resistance**; control = wire material / thickness / temperature (any one).
24. A **result** is the data you collect; a **conclusion** is what that data tells you about the prediction.

F · Physics foundations

25. A physical quantity needs a **number and a unit** — “5” could be 5 g, 5 kg...; the unit gives it meaning.
26. Scalars: **speed, distance, mass**. Vectors: **force, velocity**.
27. $\rho = m/V = 240 \div 30 = \mathbf{8\ g/cm^3}$.
28. $v = s/t = 100 \div 12.5 = \mathbf{8.0\ m/s}$.
29. Any two of: kinetic, gravitational, chemical, elastic, thermal. **Conservation**: energy cannot be created or destroyed, only stored or transferred — the total stays the same.

Created by Semra SG · for my Year 8 scientists, with all my best wishes for IGCSE.